

Rahul Rawat

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Siemens

Hiring Team

Application: Mandatory Internship, Data Science and Deep Learning for Energy Systems

Dear Hiring Team,

I am applying for the Mandatory Internship on Data Science and Deep Learning for Energy Systems at Siemens, based in Munich or Erlangen at 35 hours per week. The brief on forecasting, modeling, and optimization for energy systems using Python and deep learning frameworks maps directly to the time series and cloud data pipeline work I have been shipping over the past year.

At eRay GmbH I built an end to end recursive time series pipeline forecasting chlorophyll a, turbidity, pH, and dissolved oxygen for a German lake, benchmarking six models head to head including Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, and used CatBoost multi quantile regression to produce asymmetric 80 percent prediction intervals for decision support under uncertainty. I enforced strict anti leakage rules and surfaced the honest finding that some indicators are physically predictable while others are not without live optical sensors. That forecasting and honest evaluation discipline transfers directly to energy demand and generation modeling.

In my Real Time Flight Tracking Data Pipeline I collected live positions from the OpenSky Network every 30 seconds, enriched them against four data sources including weather, and orchestrated PySpark and dbt on Google Cloud with Apache Airflow so batch and real time layers refresh automatically every 15 minutes across more than 128 thousand records. In CreditIQ I built rigorous evaluation and benchmarking harnesses with SHAP driven analysis and unit tests at 100 percent branch coverage. That is the same automation and evaluation mindset your energy systems work needs.

I am proficient in Python and comfortable with deep learning frameworks, hold the NVIDIA Building LLM Applications With Prompt Engineering, AWS Academy Cloud Foundations, and SAS Certified Specialist Visual Business Analytics certificates, and my current German level is B1 in progress. I would be glad to start the Mandatory Internship for six months at 35 hours per week on the schedule that fits your team.

Kind regards,

Rahul Rawat